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ARTICLE

# Exposure to individual climate solutions does not reduce support for systemic solutions: a cross-cultural study in Australia and Iran

Omid Ghasemi[1], Mina Almasi[2], Mahta Khaki[3], Fateme Fallahi[4], Moslem Solhirad[5] and Ben R. Newell[1,6]

[1]UNSW Institute for Climate Risk & Response, University of New South Wales, Sydney, NSW, Australia; [2]Department of Psychology, Kermanshah Branch, Islamic Azad University, Kermanshah, Iran; [3]Department of Human Resource Management, University of Tehran, Tehran, Iran; [4]Department of Social Science, Technische Universität Kaiserslautern (RPTU), Kaiserslautern, Germany; [5]Philosophy of Science Department, Sharif University of Technology, Tehran, Iran and [6]School of Psychology, University of New South Wales, Sydney, NSW, Australia
**Corresponding author:** Omid Ghasemi; Email: o.ghasemi@unsw.edu.au

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**Abstract**

Individual-level solutions, such as lifestyle changes, are often regarded as cost-effective complements to systemic policies like government regulations in addressing climate change. However, some argue that exposure to these solutions may shift focus away from systemic interventions, potentially crowding-out support for large-scale policies. Across two experiments in Iran ($N = 303$) and Australia ($N = 308$ and 625), we tested whether exposure to different numbers of individual and systemic policies would affect individuals' perception of systemic climate solutions. Results showed that, regardless of country or exposure type, whether through daily life or experimental manipulation, participants consistently supported systemic policies and changes, attributed greater responsibility to the government, perceived minimal conflict between systemic and individual approaches, rated systemic policies as more effective than individual ones, and viewed individual actions as only moderately substitutable for systemic solutions. These findings challenge the crowding-out hypothesis, suggesting that exposure to individual solutions does not diminish support for systemic policies or structural reforms.

**Keywords:** attention opportunity cost; behavioral public policy; climate change mitigation; climate policies; crowding-out

## Introduction

To mitigate the effects of anthropogenic climate change, a range of solutions have been proposed and implemented that vary in scope, goals, methods, target behaviors, costs,

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and effectiveness. A recent trend in behavioral sciences categorizes these interventions into systemic and individual policies (Hagmann *et al.*, 2019, 2023; Chater and Loewenstein, 2022; Connolly *et al.*, 2024). Systemic policies, such as carbon taxes, typically involve government-led efforts to change the economic, legal, or infrastructural systems that shape behavior. Despite their high effectiveness, the implementation of these system-level solutions often encounters challenges, ranging from political polarizations (Newell *et al.*, 2023; Smith *et al.*, 2024) lobbying by vested interests (Oreskes and Conway, 2010; Pearson *et al.*, 2024), public skepticism and political opposition (Hornsey and Lewandowsky, 2022; Stanley *et al.*, 2024), and concerns about potential costs (Fesenfeld and Rinscheid, 2021). In response, recent years have seen an increase in focus on individual actions and solutions, designed to encourage personal actions that reduce carbon emissions by changing lifestyles and consumption patterns (Newell *et al.*, 2021).

Interventions are typically categorized as systemic when they involve structural changes through political, economic, or institutional mechanisms, such as taxes, regulations, or infrastructure investments. In contrast, individual interventions target personal behaviors like consumption and lifestyle choices through nudges, incentives, or informational campaigns (Werfel, 2017; Sparkman *et al.*, 2021; Chater and Loewenstein, 2022; Goldwert and Vlasceanu, 2025). This distinction has been described using various terms, including individualistic vs systemic (Kukowski *et al.*, 2023), demand-side vs traditional (Berger *et al.*, 2024), soft vs hard (Banerjee *et al.*, 2021), upstream vs downstream (Hampton and Whitmarsh, 2023), i-frame vs s-frame (Chater and Loewenstein, 2022), and nudge vs push (d'Adda *et al.*, 2017) interventions.

Some scholars argue that individual-level interventions can work alongside systemic policies and may even foster greater demand for structural change (Brownstein *et al.*, 2022; Constantino *et al.*, 2022; Hallsworth, 2023; Kukowski *et al.*, 2023; Goldwert and Vlasceanu, 2025). In contrast, others caution that highlighting individual interventions may shift attention away from systemic solutions and crowd-out public support for them (Chater and Loewenstein, 2022; Hagmann *et al.*, 2023; Connolly *et al.*, 2024). This study examines whether exposure to varying numbers of individual-level climate solutions reduces support for systemic policies – a phenomenon known as the crowding-out effect. Building on findings by Hagmann *et al.* (2023), which demonstrated that exposure to individual solutions can undermine support for broader systemic interventions, the current study explores this effect in Iran and Australia. These two countries differ markedly in their cultural, political, economic and climate vulnerability profiles, offering a unique cross-national test of the crowding-out effect.

### *The crowding-out effect*

The crowding-out effect occurs when individual actions and interventions reduce support for systemic policies. This phenomenon has been observed in several studies where participants, when reminded of their personal behaviors or exposed to individual-level solutions, showed decreased support for broader systemic reforms (Campbell-Arvai *et al.*, 2017; Werfel, 2017; Hagmann *et al.*, 2019; Raimi *et al.*, 2019; Knook *et al.*, 2022; Ling *et al.*, 2024; Goldwert *et al.*, 2025). For instance, Werfel (2017)

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found that completing an energy-saving checklist made participants less supportive of increasing carbon taxes.

More relevantly, across five experiments, Hagmann *et al.* (2023) showed that simply presenting participants with a set of individual interventions (i.e. individual condition) led them to generate more individual than systemic solutions when later asked to propose policies for societal challenges, compared to those who were presented with systemic policies (systemic condition). Participants in the individual condition also attributed more responsibility to individuals than to governments and rated systemic interventions as less effective across the domains of climate, finance and health. These findings were consistent with earlier studies, showing that even a simple acknowledgment of individual action or placing individual solutions alongside systemic ones can reduce support for systemic policies (Werfel, 2017; Hagmann *et al.*, 2019).

Drawing on such findings, some researchers argue that allocating attention to individual actions may trigger what is known as the *attentional opportunity cost*. This refers to the idea that limited cognitive and emotional resources are diverted away from more impactful systemic solutions. This, in turn, undermines mitigation efforts by framing individuals, rather than the systems they live in, as both the cause and the solution to the problem (Atkinson and Jacquet, 2022; Hagmann *et al.*, 2023). Supporting this idea, Sisco *et al.* (2023) observed that while concern about climate change remained stable, public and media attention to climate-related issues decreased as focus shifted to COVID-19. Relatedly, individual actions and solutions are claimed to diminish worry, concern and urgency (Knook *et al.*, 2022; but see Carrico *et al.*, 2018), create false hope for quick, effective solutions (Hagmann *et al.*, 2019; Chater and Loewenstein, 2022), create a sense of making adequate progress (Werfel, 2017), provide a moral justification for subsequent non-sustainable behaviors (Blanken *et al.*, 2015; Burger *et al.*, 2022), and promote the belief that a single action is enough when more significant or multiple measures are needed (Weber, 1997).

### *Individual actions do not undermine systemic support*

The crowding-out effect is not consistently observed across studies. Several experiments have failed to find a negative association between support for individual-level interventions, or willingness to engage in individual actions, and support for systemic policies (Cherry *et al.*, 2021; Lacroix *et al.*, 2022; Kukowski *et al.*, 2023). For instance, Lacroix *et al.* (2022) showed that engaging in individual pro-environmental behaviors did not lower support for collective actions or policies. Carrico *et al.* (2018) found that individuals who reduced red meat consumption for environmental reasons were equally likely to donate to an environmental organization and even reported increased climate concern. Similarly, Sparkman *et al.* (2021) found that while recalling past environmental actions initially reduced support for a carbon tax, this effect disappeared and even reversed when participants reflected on how these actions aligned with their values.

Individual consumption choices while shaped by broader political, economic and industrial systems are a substantial contributor to carbon emissions (Meinrenken *et al.*, 2022). As a result, many argue that successful decarbonization requires improving individual decision-making and reshaping the environments in which individuals act (Hertwig and Grüne-Yanoff, 2017; Thaler and Sunstein, 2021). The choices that

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individuals make, such as the food they eat or the products they purchase, do not occur in isolation; rather, they are both shaped by and contribute to broader systemic forces. These choices operate as signals that communicate acceptable norms within communities and influence broader systemic change. For instance, consumption patterns can shape market responses and inform policymakers about public demand, thereby generating bottom-up pressure for structural reforms (Brownstein *et al.*, 2022; Constantino *et al.*, 2022; Nielsen *et al.*, 2024). Thus, individual-level interventions aimed at shaping these behaviors, ranging from nudges and boosts to public campaigns, have become increasingly popular in recent years. They are often considered cost-effective and politically feasible tools for addressing complex societal challenges (Benartzi *et al.*, 2017), though questions remain about their overall effectiveness (DellaVigna and Linos, 2022; Maier *et al.*, 2022).

### *The current study*

In this study, we extended previous research, particularly Hagmann *et al.* (2023), to further examine the crowding-out effect. Specifically, we tested whether individuals exhibit zero-sum thinking when evaluating climate policies and whether exposure to individual solutions reduces their support for and perceived necessity of systemic climate policies. To investigate these questions, we conducted two experiments where participants were exposed to varying proportions of individual and systemic policies. Some participants were exposed predominantly or exclusively to individual solutions, others to systemic interventions, while additional conditions included balanced exposure or no policies. Across both experiments, we measured self-reported real-world exposure to different policy types, support for systemic climate policies, perceived necessity of systemic change, perceptions of government responsibility and policy conflict perceptions, among other measures. This design allowed us to test whether exposure to a greater number of individual policies than systemic ones would shift attention and support away from systemic policies and whether participants perceived systemic and individual policies as inherently conflicting.

This study was conducted in Australia and Iran. Given that most crowding-out studies have been conducted in Western societies, investigating this effect in a non-WEIRD (Western, Educated, Industrialized, Rich, Democratic; Henrich *et al.*, 2010) sample from a country with different political, economic and cultural structures than Western democracies is particularly important (Nielsen *et al.*, 2024). Specifically, countries with semi-closed political structures, low economic stability and limited financial resources, despite being more vulnerable to climate risks, may resist or be unable to support structural changes, potentially amplifying the crowding-out effect by promoting individual actions. Conversely, citizens in these countries may continue to support systemic policies and changes regardless of their engagement in individual actions. Comparing the results in Iran and Australia provides potentially valuable insights into the perceived relationship between individual and systemic solutions.

## Method summary

For Experiment 1, we recruited participants from Iran ($N = 303$, $M_{age} = 34.9$, 62% female) via social media, and Australian students ($N = 308$, $M_{age} = 19.6$, 69% female)

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**Table 1.** Examples of individual and systemic climate solutions used in the experimental tasks of Experiments 1 and 2. Each participant was randomly presented with a subset of these interventions depending on their experimental condition. The full list of policies and complete materials are available in the Supplementary Materials

| Systemic solutions | Individualistic solutions |
|---|---|
| Governments can implement a carbon tax for industries as a measure to combat climate change. By enforcing this tax policy, governments compel industries and factories to reduce their contribution to greenhouse gas emissions and mitigate their negative impact on climate change. | Individuals can reduce greenhouse gas emissions by consuming less meat in their meals. By reducing meat consumption, the conversion of forest lands to pasture lands decreases. |
| Non-standard construction leads to energy waste for heating and cooling in buildings. Governments can require constructors to adhere to laws and standards to prevent this energy waste. | By participating in educational campaigns, people can become more aware of the harmful effects of carbon dioxide emissions on climate change, encouraging them to take action to reduce these emissions and other pollutants. |
| The government can increase investment in solar and wind energy projects to reduce reliance on fossil fuels and transition to a cleaner energy grid. | Individuals can choose a 100% renewable electricity plan from their energy provider to support the growth of clean energy. |
| The government can increase investment in high-speed rail and expand public transport networks, reducing reliance on cars and lowering emissions. | Individuals can use public transport or cycle instead of driving whenever possible to cut personal emissions. |
| The government can mandate energy efficiency standards for appliances, ensuring that household and commercial devices consume less electricity. | Individuals can replace old appliances with energy-efficient models over time when they need upgrades. |

from the UNSW undergraduate pool. For Experiment 2, which was pre-registered (https://osf.io/h7kae), we recruited the Australian general population ($N = 624$, $M_{age} = 48.4$, 55% female) via Dynata. The final sample sizes reflect exclusions of those who failed more than one attention check or were under 18.

Both experiments manipulated the proportion of individual (e.g. lifestyle changes) and systemic (e.g. governmental regulations) climate policies participants were exposed to. For Experiment 1, participants were randomly assigned to one of three conditions in a between-subjects design. They evaluated 10 policies aimed at reducing carbon emissions, with conditions differing in the proportion of systemic and individual solutions presented. The systemic condition included eight systemic and two individual solutions, the individual condition included eight individual and two systemic solutions, and the control condition included an equal number of both types. Table 1 provides sample items from the set of individual and systemic interventions presented to participants during the experiments. After rating each policy's effectiveness on a 5-point scale, participants completed a series of surveys.

Participants began by evaluating their perceptions of individual and systemic policies, indicating which type they encountered more frequently in daily life or on social

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media, which they found most effective, which type they believed scientists and policymakers consider more effective in addressing climate change, and whether they viewed them as complementary or conflicting. Higher ratings corresponded to greater exposure to individual interventions, stronger perceived effectiveness of systemic solutions, and greater perceived conflict. Next, participants assessed responsibility attribution by rating the extent to which the government and individuals are responsible for addressing climate change. A difference score was calculated by subtracting the individual responsibility rating from the government responsibility rating, with higher values indicating a stronger belief in government responsibility. Following this, participants rated their support for national government climate policies, evaluating various systemic interventions such as carbon taxes and public transport investments. They also assessed the perceived need for systemic changes, indicating the extent to which major societal, political, and economic shifts are necessary to address climate change. Finally, participants completed additional measures assessing their perceived effectiveness of individual actions and their willingness to adopt pro-environmental behaviors (behavioral plasticity; Kukowski *et al.*, 2023), with subscales distinguishing high-impact and low-impact behaviors.

In Experiment 2, we retained the core experimental design of Experiment 1 but introduced several refinements to strengthen the manipulation and provide the best opportunity for detecting a potential crowding-out effect. To align more closely with prior research (e.g. Hagmann *et al.*, 2023), we adjusted the experimental conditions so that participants in the systemic condition evaluated exclusively systemic solutions (eight policies), and those in the individual condition evaluated exclusively individual solutions (eight policies). This ensured that participants were only exposed to a single type of intervention, providing a stronger test of whether individual solutions undermine support for systemic policies. Additionally, all interventions focused on energy-related policies, as we expected that if crowding-out occurs, it is more likely to manifest when participants evaluate an individual action (e.g. choosing a renewable electricity plan) that directly overlaps with a systemic alternative (e.g. electric vehicle rebates), rather than an unrelated policy (e.g. a meat tax). Thus, we added additional items to the policy support scale to distinguish between energy-related and energy-unrelated policies, allowing for exploratory analyses to determine whether exposure to individual vs systemic interventions influenced support for related vs. unrelated policies differently.

To further refine the manipulations, Experiment 2 implemented a different control condition where participants completed a science-related quiz rather than evaluating a balanced number of systemic and individual solutions. Additionally, in the policy perception section, participants responded to a systemic substitutability question, indicating the extent to which they thought individual interventions lessen the necessity of systemic policies, where higher scores represented greater perceived substitutability. Finally, given the logistical difficulties of additional cross-national data collection, Experiment 2 was carried out exclusively in Australia. Figure 1 illustrates the experimental procedure, including the manipulations and the order of surveys for both experiments. A detailed description of the sample, power analyses, methods and materials is provided in the Supplementary Materials.

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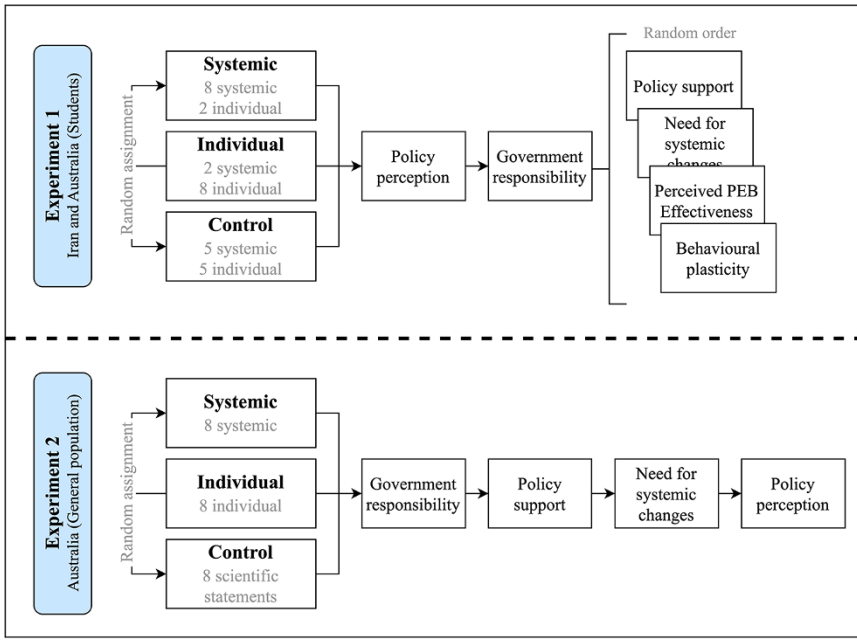



**Figure 1.** Experimental procedure for Experiments 1 and 2. Participants were randomly assigned to different conditions where they evaluated either individual or systemic climate policies. In Experiment 1, the control condition involved evaluating a mix of both individual and systemic policies, whereas in Experiment 2, the control condition required participants to complete a science-related quiz instead of evaluating climate policies. All participants then completed measures assessing policy support, perceived necessity of systemic change, responsibility attribution, and perceived policy alignment or conflict. The order of some measures differed slightly between the two experiments.

## Results

Table 2 presents key descriptive results from both experiments. Significant variations were observed across study samples, with pairwise comparisons shown in Figure S2. Iranians and Australian students reported greater exposure to individual policies than the Australian general population ($ps < 0.001$), who instead reported encountering a balanced mix of systemic and individual policies in daily life. In ratings of policies in the experiment block, systemic policies were seen as more effective across all samples, with Iranians and Australian students rating them higher than the Australian general population ($ps < 0.011$).

Perceived necessity for systemic change and climate policy support were high across all samples, with Iranians giving the highest ratings, followed by Australian students, and then the general Australian population ($ps < 0.001$). Australians reported moderately low perceived conflict between systemic and individual policies, while Iranians reported very low conflict. Iranians saw less conflict than both Australian samples ($ps < 0.001$), and students reported less than the general Australian population ($p < 0.001$). In all three samples, governments were rated as more responsible than individuals in addressing climate change ($ps < 0.001$). This difference was largest in

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**Table 2.** Average ratings of key variables for individuals, categorized by samples and experimental groups. In Experiment 1, participants in the systemic and individual groups evaluated mostly systemic or individual climate policies, respectively, while the control group saw a balanced mix of both. In Experiment 2, participants evaluated only systemic or only individual energy-related policies or completed a science quiz in the control condition

| Experiment | Group | Daily encounter [individual] | Systemic effectiveness | Need systemic changes | Policy support | Policies conflict | Government responsible | Individuals responsible | Systemic substitutability |
|---|---|---|---|---|---|---|---|---|---|
| **Experiment 1** (Iranian sample) | Control | 65.11 | 61.64 | 6.58 | 5.67 | 1.76 | 6.18 | 4.31 | |
| | Individual | 65.92 | 58.43 | 6.48 | 5.42 | 1.78 | 6.17 | 4.32 | |
| | Systemic | 63.28 | 61.52 | 6.57 | 5.40 | 1.80 | 6.03 | 4.25 | |
| | **Overall** | **64.77** | **60.56** | **6.54** | **5.50** | **1.78** | **6.13** | **4.29** | |
| **Experiment 1** (Australian students) | Control | 67.74 | 71.04 | 6.14 | 4.99 | 2.32 | 6.26 | 4.78 | |
| | Individual | 71.05 | 68.32 | 6.12 | 5.11 | 2.47 | 5.82 | 4.90 | |
| | Systemic | 61.30 | 71.26 | 6.20 | 4.98 | 2.41 | 6.14 | 4.76 | |
| | **Overall** | **66.68** | **70.21** | **6.15** | **5.03** | **2.40** | **6.08** | **4.81** | |
| **Experiment 2** (Australian general population) | Control | 45.90 | 60.87 | 5.39 | 4.68 | 2.71 | 5.41 | 4.97 | 2.87 |
| | Individual | 46.79 | 61.42 | 5.32 | 4.55 | 2.73 | 5.42 | 4.90 | 2.90 |
| | Systemic | 46.21 | 61.63 | 5.27 | 4.60 | 2.78 | 5.37 | 4.92 | 3.00 |
| | **Overall** | **46.29** | **61.3** | **5.33** | **4.62** | **2.74** | **5.40** | **4.93** | **2.92** |
| **Scale** | | 0–100 | 0–100 | 1–7 | 1–7 | 1–5 | 1–7 | 1–7 | 1–5 |

Iran, followed by Australian students, and smallest in the Australian general population ($ps < 0.007$). Finally, the Australian general population reported moderate systemic substitutability ratings, suggesting that individual solutions 'somewhat' reduce the need for systemic solutions. The correlations between the key variables measured in each experiment are presented in Tables S1, S2 and S3 of the Supplementary Materials.

### *The impact of experimental exposure to individual solutions*

During the experimental phase, participants were presented with a varying number of systemic and individual solutions and were asked to rate the effectiveness of each solution (see the percentage of effectiveness ratings for each policy in Figure S1). Experiment 1 results showed that systemic policies were rated as significantly more effective than individual policies among Iranian ($M = 3.96$ vs 3.46), $t$ (302) = 12.33, $p < 0.001$, $d = 0.71$, and Australian students ($M = 3.58$ vs 2.92), $t$ (307) = 14.12, $p < 0.001$, $d = 0.92$. This pattern held across all three experimental groups ($ps < 0.001$). However, in Experiment 2 (Australian general population), no significant difference was observed between systemic and individual policy effectiveness ratings ($M = 3.06$ vs 3.0, $p = 0.55$).

To what extent does varying the number of individual and systemic policies presented influence attitudes toward systemic policies, perceptions of systemic change necessity, responsibility attribution, policy conflict, and the substitutability of systemic solutions? As shown in Table 2, participants in both countries expressed strong support for systemic policies, recognized the need for systemic changes, attributed greater responsibility to the government than individuals, and perceived minimal conflict between systemic and individual policies, though this effect was weaker in the Australian samples. Crucially, as illustrated in Figure 2, these attitudes remained unchanged regardless of the number of individual policies participants encountered in the experiment. Across all samples, the experimental group was not a significant predictor of any key dependent variable ($ps > 0.071$, $\eta^2_{ps} < 0.018$; Detailed model results are available in Table S4 in the Supplementary Materials).

### *The impact of daily life exposure to individual solutions*

To examine how self-reported exposure to individual policies in daily life influences attitudes toward systemic policies, we conducted a series of linear regressions for each key variable. For the Australian general population in Experiment 2, the models included only exposure to individual policies as predictors. In Experiment 1, for both Iranian and Australian student samples, we also included plasticity indexes and perceived PEB effectiveness as additional predictors. To facilitate the comparison of effect sizes across predictors with different scales (e.g. 7-point and 100-point scales), we standardized all numeric predictors to $z$-scores.

As Figure 3 shows, daily exposure to individual interventions was associated with lower climate policy support ($\beta = -0.13$, $p = 0.006$) in the Australian general population but did not negatively predict attitudes toward systemic policies or changes in any sample. Instead, greater exposure to individual solutions was associated with a stronger perceived need for systemic changes among Australian students

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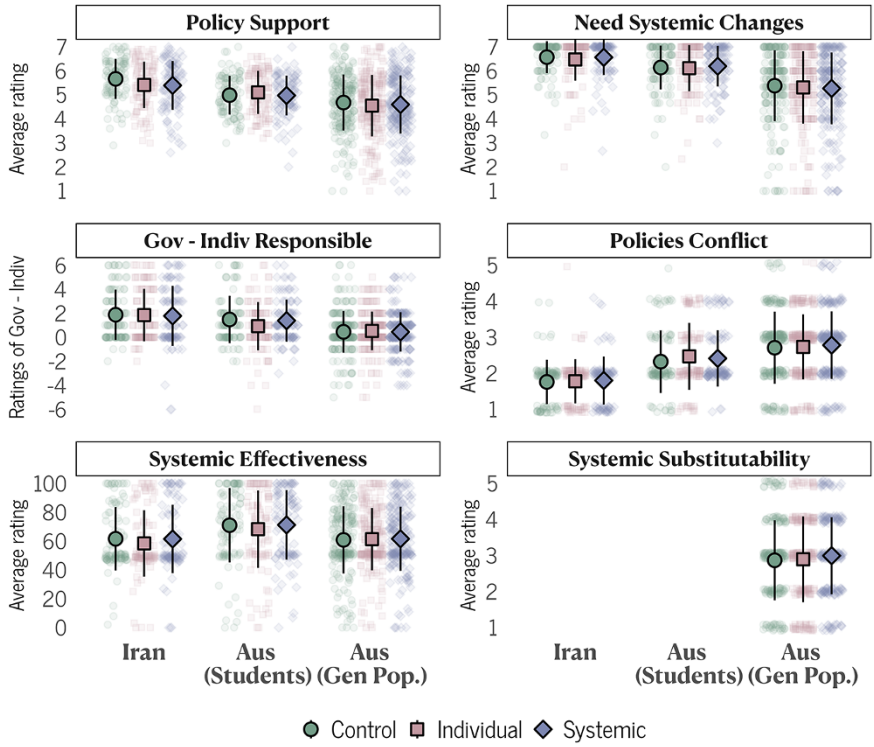



**Figure 2.** Participants' average and raw ratings for policy support, the need for systemic changes, the perceived onus of responsibility on the government, the perceived contradiction between systemic and individual policies, and the perceived substitutability of systemic solutions across experimental groups and samples. Group assignments reflect exposure to either mostly systemic, mostly individual, or (in control conditions) a balanced policy set (Experiment 1) or a science quiz (Experiment 2). Error bars represent standard deviations.

($\beta = 0.17$, $p < 0.001$), higher attribution of responsibility to the government in Iran ($\beta = 0.45$, $p < 0.001$), lower perceived conflict between systemic and individual solutions ($\beta = -0.11$, $p = 0.002$), and reduced belief that systemic policies can be replaced by individual solutions in the Australian general population ($\beta = -0.18$, $p < 0.001$). No other effects reached statistical significance (see Tables S5, S6 and S7 for the full regression model outputs).

Next, we examined additional predictors to determine whether individual actions and beliefs about their effectiveness were associated with support for systemic policies and perceptions of their necessity. The results indicated that low-impact behavioral plasticity was positively linked to policy support ($\beta = 0.12$, $p = 0.014$) and the perceived necessity of systemic policies ($\beta = 0.18$, $p < 0.001$) among Australian students, as well as greater perceived effectiveness of systemic policies in Iran ($\beta = 3.01$, $p < 0.044$). In contrast, high-impact behavioral plasticity was associated with stronger climate policy support in both countries (Iran: $\beta = 0.36$, $p < 0.001$; Australian students: $\beta = 0.27$, $p < 0.001$), greater perceived need for systemic change in Iran ($\beta = .13$, $p = 0.016$),

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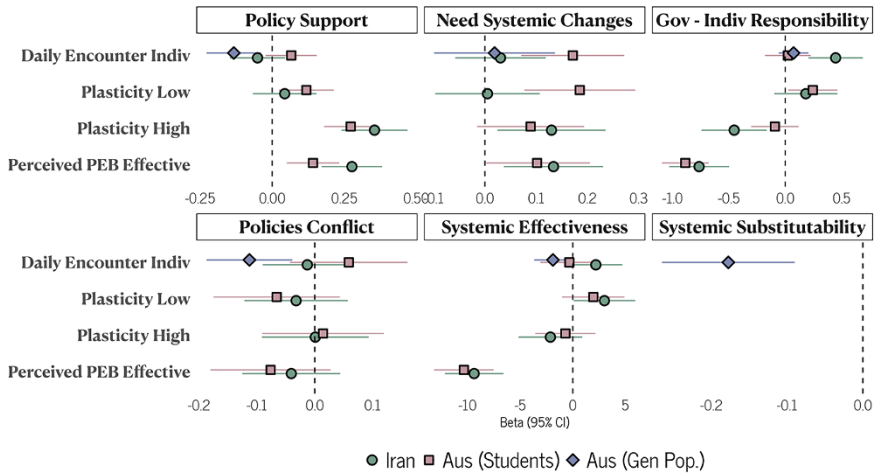



**Figure 3.** Regression coefficients for models predicting policy support, the need for systemic changes, the relative responsibility of government compared to individuals, and the perceived conflict between systemic and individual policies for all three samples. All numeric predictors are *z*-transformed and are displayed on the *x*-axis. The horizontal error bars represent the 95% confidence intervals for each coefficient.

and a shift in responsibility attribution toward individuals relative to governments in Iran ($\beta = -0.45$, $p = 0.002$).

Greater perceived effectiveness of individual actions was linked to increased policy support in both countries (Iran: $\beta = 0.28$, $p < 0.001$; Australian students: $\beta = 0.14$, $p = 0.002$), as well as a stronger perceived need for structural changes in Iran ($\beta = .13$, $p = .007$). However, it also correlated with reduced government responsibility attribution (Iran: $\beta = -0.77$, $p < 0.001$; Australian students: $\beta = -0.89$, $p < 0.001$), and lower perceived effectiveness of systemic policies (Iran: $\beta = -9.39$, $p < 0.001$; Australian students: $\beta = -10.36$, $p < 0.001$) in both countries. Together, these findings suggest that while a greater willingness to adopt sustainable behaviors and stronger beliefs in their effectiveness are linked to increased support for climate policies and the perceived need for systemic change, they also correspond with attributing more responsibility to individuals rather than governments and a diminished belief in the effectiveness of systemic policies.

### *Additional analyses*

In Experiment 1, participants were asked which type of policy (systemic or individual) they believed scientists and policymakers consider more effective in addressing climate change (0 = individual interventions, 100 = systemic interventions). The results showed that Australian students more strongly believed that both scientists ($M_{\text{Aus}} = 70.33$ vs $M_{\text{Iran}} = 61.09$) and policymakers ($M_{\text{Aus}} = 61.66$ vs $M_{\text{Iran}} = 44.48$) favor systemic solutions compared to the Iranian sample. In fact, Iranian participants on average believed that policymakers slightly favor individual policies. The experimental manipulation of exposure to different numbers of systemic and individual

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policies did not significantly predict perceptions of scientists' or policymakers' views in either country ($ps > 0.054$). Similarly, self-reported real-world exposure to individual interventions did not predict these perceptions among Australian participants ($ps > 0.101$). However, in the Iranian sample, greater reported exposure to individual solutions was associated with stronger belief that scientists endorse systemic policies ($\beta = 2.74$, $p = 0.041$), but weaker belief that policymakers favor systemic approaches ($\beta = -6.76$, $p < 0.001$).

Moreover, immediately after the experimental block in Experiment 1, participants answered a manipulation check question about which category (individual vs systemic) most of the presented policies belonged to. The results showed that participants in the systemic condition reported seeing more systemic policies compared to the other two groups, and participants in the control group reported seeing more systemic policies compared to the individual condition ($ps < 0.009$). These results validate the effectiveness of the experimental manipulation and demonstrate participants' awareness of the base rates for each policy category.

In Experiment 2, participants rated their support for a series of climate policies that included both energy-related and unrelated interventions to explore whether exposure to different policy types would influence support for specific categories. As Figure S3 shows, the experimental manipulation did not significantly predict support for either energy-related or energy-unrelated policies in Experiment 2 ($ps > 0.392$).

## Discussion

While individual actions are often seen as valuable co-solutions to climate change (Constantino *et al.*, 2022; Bradley *et al.*, 2024; Nielsen *et al.*, 2024), some researchers argue they may inadvertently undermine systemic change by shifting focus away from more impactful policies (Werfel, 2017; Hagmann *et al.*, 2019, 2023; Chater and Loewenstein, 2022). This study tested this claim across two culturally and politically distinct countries, examining whether exposure to individual solutions, both experimentally manipulated and encountered in daily life, predicts lower support for and perceived necessity of systemic policies. Furthermore, we examined the relationship between individual pro-environmental behaviors and support for systemic solutions.

The findings showed that, in all three samples, exposure to varying numbers of individual interventions through experimental manipulation did not reduce support for systemic climate policies, perceived need for structural change, beliefs about government responsibility, perceived alignment between individual and systemic policies, or the view that systemic solutions cannot be replaced by individual actions. Higher daily exposure to individual solutions was generally unrelated or positively associated with these measures, except among the Australian general population, where it was negatively linked to policy support and perceived effectiveness of systemic interventions. Additionally, higher willingness to engage in eco-friendly behaviors, especially those with greater impact, and stronger beliefs in their effectiveness were linked to greater support for systemic solutions and a stronger perceived need for structural change. At the same time, these factors were also associated with shifting more responsibility to individuals rather than governments and perceiving systemic solutions as less effective.

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Across both experiments, participants showed little evidence of zero-sum thinking and exposure to individual solutions did not reduce their support for systemic policies. These results align with prior research indicating that the crowding-out effect is not universal and does not consistently emerge across contexts or manipulations (Carrico *et al.*, 2018; Cherry *et al.*, 2021; Sparkman *et al.*, 2021; Lacroix *et al.*, 2022; Kukowski *et al.*, 2023). At the same time, the findings largely challenge the crowding-out hypothesis, which suggests that promoting individual solutions diverts attention from systemic policies. Two key factors may explain this discrepancy. First, our experimental design differed from these previous studies. For instance, in Hagmann *et al.* (2023), participants not only rated individual and systemic policies but were also asked to generate their own policy ideas before assessing government responsibility and the importance of each policy type. This additional policy-generation task may have prompted participants to generate policies similar to those that they had just encountered. Second, sample differences may have played a role. While Werfel (2017) recruited Japanese participants and Hagmann *et al.* (2019, 2023) studied U.S. participants, our samples from Australia and Iran may have exhibited weaker tendencies toward zero-sum thinking.

The results of this study revealed both similarities and differences across samples and countries. Australian students and Iranians perceived minimal conflict between individual and systemic policies, viewing them as complementary, and the general Australian sample reported a moderately low level of conflict, though still below the midpoint. In both countries, governments were seen as more responsible than individuals for addressing climate change with the difference being largest among the Iranian participants. Across all three samples, participants expressed moderately high support for climate policies, rated them as effective, and perceived them as necessary. Despite cultural and political differences, support for systemic policies and structural changes remained consistent, regardless of participants' exposure to individual solutions, either experimentally or in daily life.

Examining the impact of individual actions on systemic changes is both theoretically and practically significant. If focusing on individual actions diverts attention and support from systemic changes, efforts should prioritize more impactful systemic policies. Conversely, if crowding-out does not always occur, policymakers can consider individual actions as complementary rather than contradictory to systemic approaches. In such cases, individual interventions can shift behaviors toward sustainability, particularly where structural changes face obstacles such as political polarization, rigid systems, public disengagement, or distrust (Drews and van den Bergh, 2016; Newell *et al.*, 2023; Bretter and Schulz, 2025; Ghasemi *et al.*, 2025b). Furthermore, as individuals often underestimate the willingness of others to participate in pro-environmental behaviors (Andre *et al.*, 2024), engaging in more individual actions can act as a social signal, reducing this 'pluralistic ignorance' and fostering bottom-up pressure for systemic change. Lifestyle changes, when combined with actions like collective advocacy, financial investments, and political support, can establish social norms that significantly enhance mitigation efforts (Constantino *et al.*, 2022; Nielsen *et al.*, 2024). Individual interventions can also be designed to increase acceptance of system-level policies and facilitate structural changes (Brownstein *et al.*, 2022; Hallsworth, 2023). In this view, behavioral science serves as a valuable tool for designing and implementing interventions that complement, rather than replace, system-level policies.

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While these findings highlight the value of individual interventions for climate mitigation, it is critical to recognize their limitations and communicate their effectiveness truthfully. Some pro-environmental behaviors and interventions are less impactful than other solutions in addressing climate change (DellaVigna and Linos, 2022; Maier *et al.*, 2022; Berger *et al.*, 2024), and framing these solutions as substitutes for systemic policies or exaggerating their effectiveness could actually crowd-out support for broader climate policies (Hagmann *et al.*, 2019; Raimi *et al.*, 2019; Atkinson and Jacquet, 2022). Furthermore, the design of interventions may benefit from a reassessment of the assumption that human decision-making is inherently irrational. Traditional frameworks often leverage cognitive biases, such as preferences for default options or framing effects, to nudge individuals toward socially beneficial outcomes (Thaler and Sunstein, 2021). However, recent studies suggest that individuals actively interpret interventions and the intentions behind them, rather than passively reacting (Madsen *et al.*, 2024; Szollosi *et al.*, 2025; Ghasemi *et al.*, 2025a). Thus, individual interventions should be seen not as exploiting cognitive biases, but as complementary tools that work symbiotically with systemic changes to assist rational citizens in making informed decisions.

The distinction between individual and systemic interventions is frequently used in both academic debates and policymaking. However, the criteria for categorizing interventions into these two types are neither clear-cut nor universally accepted (Brownstein *et al.*, 2022; Hallsworth, 2023; Nielsen *et al.*, 2024; Goldwert and Vlasceanu, 2025). Systemic interventions are typically characterized by their focus on structural change, greater potential effectiveness, and higher implementation costs. Individual interventions, by contrast, are generally defined as targeting individual behavior, involving relatively lower costs, and often having a more modest impact. While some researchers have proposed more structured distinctions (for instance, describing individual policies as those aimed at changing the behavior of players, and systemic policies as those that seek to change the rules of the game; Chater and Loewenstein, 2022; Connolly *et al.*, 2024), none of these criteria are definitive. Many real-world policies contain elements of both categories, making them difficult to classify neatly (Brownstein *et al.*, 2022).

Furthermore, we do not argue that maintaining this dichotomy is inherently beneficial. This distinction can be useful when it highlights how powerful actors sometimes deflect responsibility onto individuals for systemic issues, as seen in campaigns that overemphasize personal carbon footprints while downplaying corporate responsibility (for examples, see Chater and Loewenstein, 2022). However, drawing a strict line between individual and systemic actions might encourage oppositional or zero-sum thinking, which could potentially impede climate mitigation efforts by promoting a contradictory view that supporting one type of intervention undermines the other (Brownstein *et al.*, 2022). It can lead to the mistaken belief that one kind of policy is sufficient, whereas the success of systemic solutions often hinges on individual trust, support, and behavior (Drews and van den Bergh, 2016; Hallsworth, 2023), and individual change can be effectively enabled or sustained by structural reform (Connolly *et al.*, 2024; Nielsen *et al.*, 2024).

A key limitation of this study is its cross-sectional design, which prevents assessing shifts in climate policy support after long-term exposure to individual interventions.

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Future research could examine the impact of individual-targeted interventions at multiple time points. Additionally, while this study included two culturally distinct countries, broader cross-cultural comparisons, particularly with nations differing in political stability and economic development, could improve the generalizability of the findings. Larger and more representative samples would also enhance the accuracy of parameter estimates. A further potentially interesting direction for future research would be to vary the effectiveness of individual or systemic interventions to see how this shapes support for other policies. This may also shed light on potential zero-sum thinking not only among the public but also among policymakers. For example, future studies could examine how bundling policies with varying levels of effectiveness influences the support for and implementation of more effective climate policies (Hagmann *et al.*, 2019; Marshall *et al.*, 2024). Finally, future studies could refine both the experimental manipulations and measures of policy exposure, potentially using real-world data on actual exposure to individual versus systemic interventions.

## Conclusion

Behavioral interventions targeting individuals are often seen as valuable solutions to societal challenges, though some argue they may crowd out support for more effective systemic interventions. This study provides evidence against the crowding-out effect, showing that individuals maintain support for systemic policies even when exposed to multiple individual actions. These findings highlight the importance of integrating individual and systemic approaches to achieve meaningful and sustained climate action.

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